Statistical Data Analysis and Visualization Lab

Prompts to build the interactive web application

Use these prompts one at a time in your preferred AI coding or vibe-coding tool. Build the application incrementally and test each module before proceeding.

Prompt 1 — Create the basic application

Create an interactive educational web application titled **"Continuous Probability Distributions: Industry Applications"** for undergraduate AIML students.

Build a clean, responsive interface with a home page and three modules:

1. Normal Distribution — Manufacturing Quality Control
2. Exponential Distribution — Cloud Request Inter-arrival Time
3. Central Limit Theorem — Warehouse Processing Time
   Use simple language suitable for beginners. Each module must include:

- Industry problem statement
- Input parameters
- Mathematical formula
- Interactive calculation section
- Dynamic visualization
- Interpretation of results

Use Python only if the chosen framework supports it; otherwise use the framework's native numerical and charting libraries. Implement the application using the existing project setup and do not introduce unnecessary dependencies.

For now, create the navigation and page layouts. Do not implement all calculations yet.

Output:

ProbabilityLab

Home | Normal Distribution | Exponential Distribution | Central Limit Theorem

Continuous Probability Distributions: Industry Applications

Learn how probability distributions are used in real-world AIML and industry problems such as manufacturing quality, cloud computing, and warehouse operations.

Start Learning →

Explore the Modules

Understand important probability concepts through simple industry-based examples and interactive learning activities.

Explore the Modules

Understand important probability concepts through simple industry-based examples and interactive learning activities.

Normal Distribution

Learn how manufacturing companies use the normal distribution to monitor product quality and identify unusual measurements.

Open Module →

Exponential Distribution

Understand how cloud systems can model the time between incoming requests using the exponential distribution.

Open Module →

Central Limit Theorem

Explore how warehouse managers can use sample averages to understand processing time and make better decisions.

Open Module →

ProbabilityLab

Home | Normal Distribution | Exponential Distribution | Central Limit Theorem

Normal Distribution

Manufacturing Quality Control

Industry Problem

A manufacturing company produces metal components. The diameter of each component is expected to be close to a target value. Small variations are normal, but very large variations may indicate a quality problem.

The normal distribution can help engineers understand how the component measurements are distributed around the average value.

Input Parameters

Mean (μ)

50

Standard Deviation (σ)

2

Measurement (x)

53

Calculate

Mathematical Formula

$$f(x) = 1 / (\sigma\sqrt{(2\pi)}) \times e^{-(x-\mu)^2 / 2\sigma^2}$$

The formula describes the probability density of a normally distributed variable.

Dynamic Visualization



Normal Distribution Graph

Interactive bell-curve visualization will be added in the calculation stage.

Interpretation of Results

Result Interpretation

After calculation is implemented, this section will explain whether the selected measurement is close to the average or unusually far from it.

Exponential Distribution

Cloud Request Inter-arrival Time

Industry Problem

A cloud application receives requests from users. The time between two consecutive requests can vary.

Cloud engineers can use the exponential distribution to model the waiting time between independent requests arriving at a system.

Input Parameters

Average Arrival Rate (λ)

5

Waiting Time (t)

0.5

Calculate

Mathematical Formula

$$f(t) = \lambda e^{-\lambda t}$$

$$P(T > t) = e^{-\lambda t}$$

The exponential distribution models waiting time between events.

Dynamic Visualization



Exponential Distribution Curve

Interactive waiting-time curve will be added in the calculation stage.

Interpretation of Results

Result Interpretation

After calculation is implemented, this section will explain the probability of a cloud request arriving within the selected time interval.

Central Limit Theorem

Warehouse Processing Time

📦 Industry Problem

A warehouse processes many customer orders every day. The processing time of individual orders can vary depending on the order size, product location, and number of items.

Warehouse managers can use the Central Limit Theorem to understand the behavior of the average processing time for a sample of orders.

Input Parameters

Population Mean (μ)

30

Population Standard Deviation (σ)

8

Sample Size (n)

36

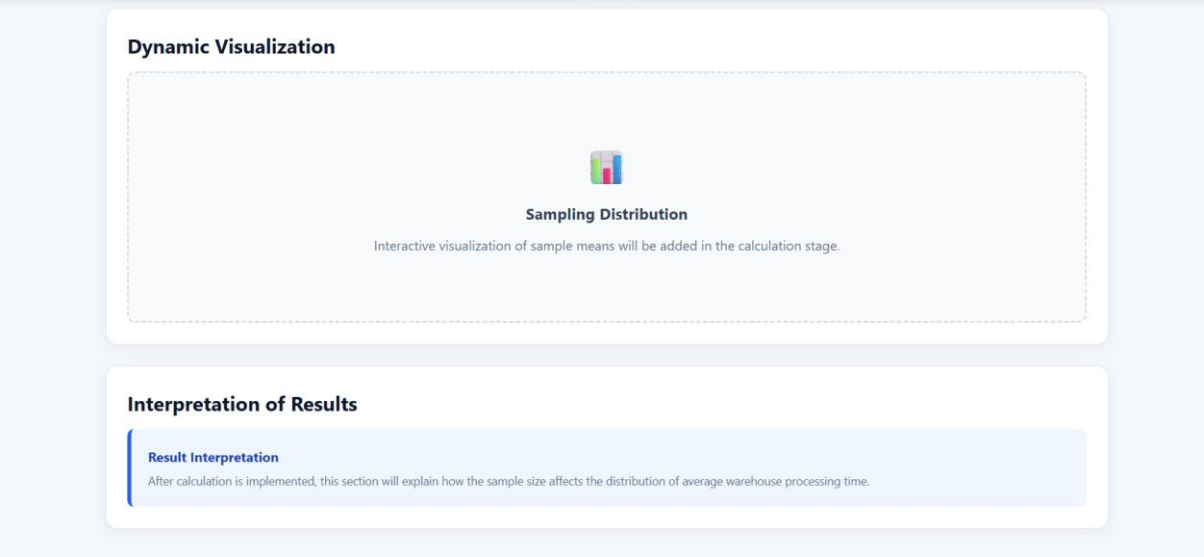
Calculate

Mathematical Formula

$$\mu_{\bar{x}} = \mu$$

$$\sigma_{\bar{x}} = \sigma / \sqrt{n}$$

As the sample size becomes larger, the distribution of sample means approaches a normal distribution.



Prompt 2 — Implement the Normal Distribution module

Implement the **Normal Distribution — Manufacturing Quality Control** module.

Use this illustrative example:

- Target shaft diameter: 10 mm
- Mean diameter $\mu = 10$ mm
- Standard deviation $\sigma = 0.05$ mm
- Lower specification limit = 9.90 mm
- Upper specification limit = 10.10 mm

Provide interactive inputs for μ, σ, lower limit, and upper limit.

Calculate:

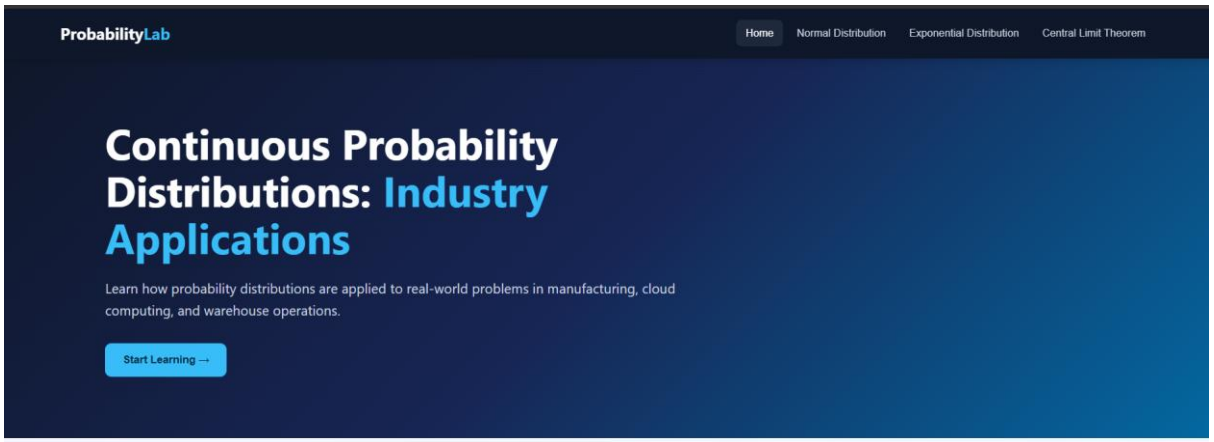
1. Probability density at a selected diameter.
2. Probability that diameter lies between the lower and upper specification limits.
3. Probability of a diameter below the lower limit.
4. Probability of a diameter above the upper limit.
5. Expected percentage of defective products.

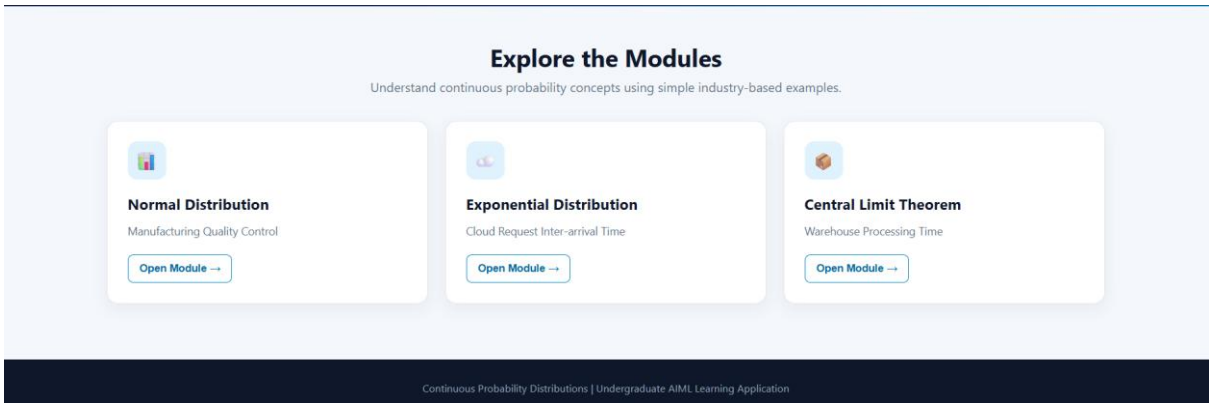
Use the Normal CDF to calculate interval probabilities. Display a dynamic bell-shaped density curve and shade the acceptable region and the two defective regions in different colors.

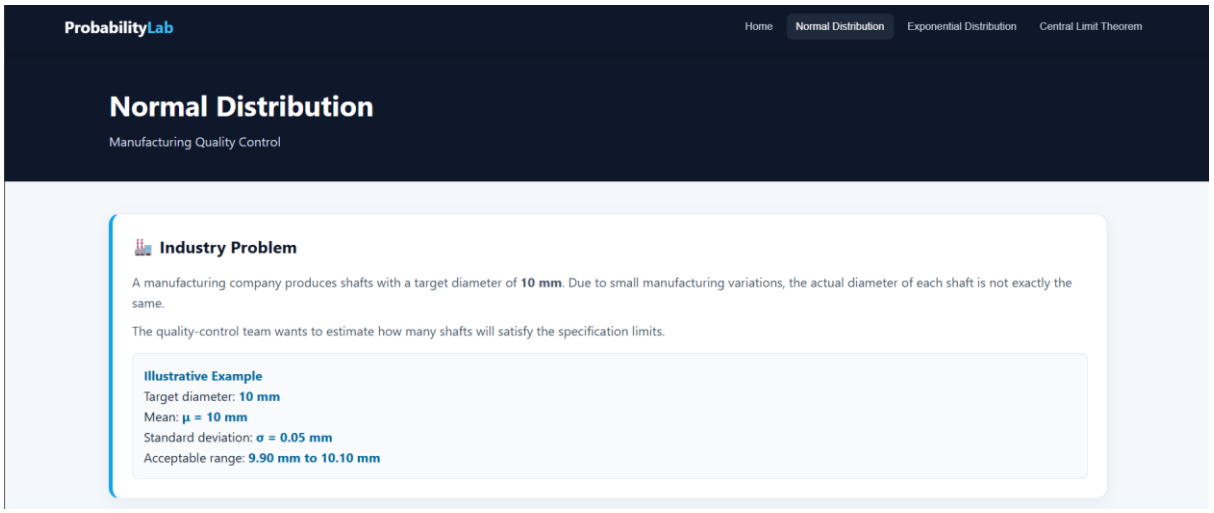
When any input changes, update the calculations and graph immediately. Validate that σ is positive and the lower limit is less than the upper limit.

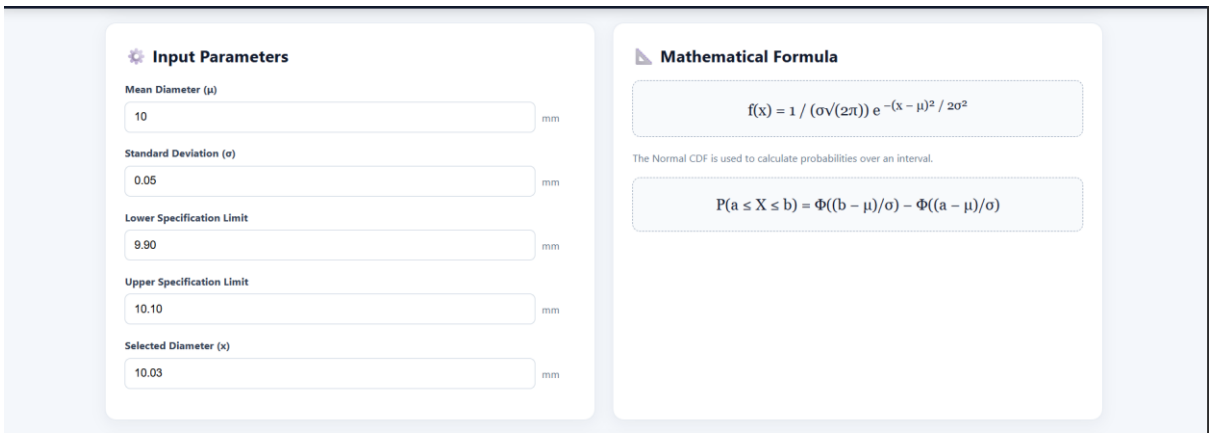
Display formulas, numerical answers, units, and a plain-language quality-control interpretation.

Output:


ProbabilityLab
Home
Normal Distribution
Exponential Distribution
Central Limit Theorem
Continuous Probability Distributions: Industry Applications
Learn how probability distributions are applied to real-world problems in manufacturing, cloud computing, and warehouse operations.
Start Learning →


Explore the Modules
Understand continuous probability concepts using simple industry-based examples.
Normal Distribution
Manufacturing Quality Control
Open Module →
Exponential Distribution
Cloud Request Inter-arrival Time
Open Module →
Central Limit Theorem
Warehouse Processing Time
Open Module →
Continuous Probability Distributions | Undergraduate AIML Learning Application


ProbabilityLab
Home
Normal Distribution
Exponential Distribution
Central Limit Theorem
Normal Distribution
Manufacturing Quality Control
Industry Problem
A manufacturing company produces shafts with a target diameter of 10 mm. Due to small manufacturing variations, the actual diameter of each shaft is not exactly the same.
The quality-control team wants to estimate how many shafts will satisfy the specification limits.
Illustrative Example
Target diameter: 10 mm
Mean: μ = 10 mm
Standard deviation: σ = 0.05 mm
Acceptable range: 9.90 mm to 10.10 mm


Input Parameters
Mean Diameter (μ)
10
mm
Standard Deviation (σ)
0.05
mm
Lower Specification Limit
9.90
mm
Upper Specification Limit
10.10
mm
Selected Diameter (x)
10.03
mm
Mathematical Formula
f(x) = 1 / (σ√(2π)) e^−(x − μ)² / 2σ²
The Normal CDF is used to calculate probabilities over an interval.
P(a ≤ X ≤ b) = Φ((b − μ)/σ) − Φ((a − μ)/σ)

Calculated Results

Density at selected diameter **6.6645** probability density per mm	Probability within specification **95.4500%** P(9.90 ≤ X ≤ 10.10)
Below lower limit **2.2750%** P(X < lower limit)	Above upper limit **2.2750%** P(X > upper limit)
Expected defective products **4.5500%** below + above specification	Selected diameter **10.030 mm** current x value

Dynamic Normal Distribution

The graph updates automatically when any input changes. Green represents acceptable production, while orange and red represent defective regions.

Quality-Control Interpretation

What does this mean?

Approximately **95.450%** of shafts are expected to meet the specification limits of **9.900 mm** to **10.100 mm**.

The expected defective percentage is **4.550%**. Of this defective production, approximately **2.275%** is expected to be below the lower limit and **2.275%** above the upper limit.

The process has a moderate level of expected defects. The manufacturing team should monitor the variation and consider process improvement if this defect rate is higher than the company's target.

Beginner Explanation

Normal Distribution: The bell-shaped curve represents how shaft diameters are expected to vary around the average diameter.

Specification limits: Shafts between the lower and upper limits are considered acceptable. Shafts outside these limits are treated as defective.

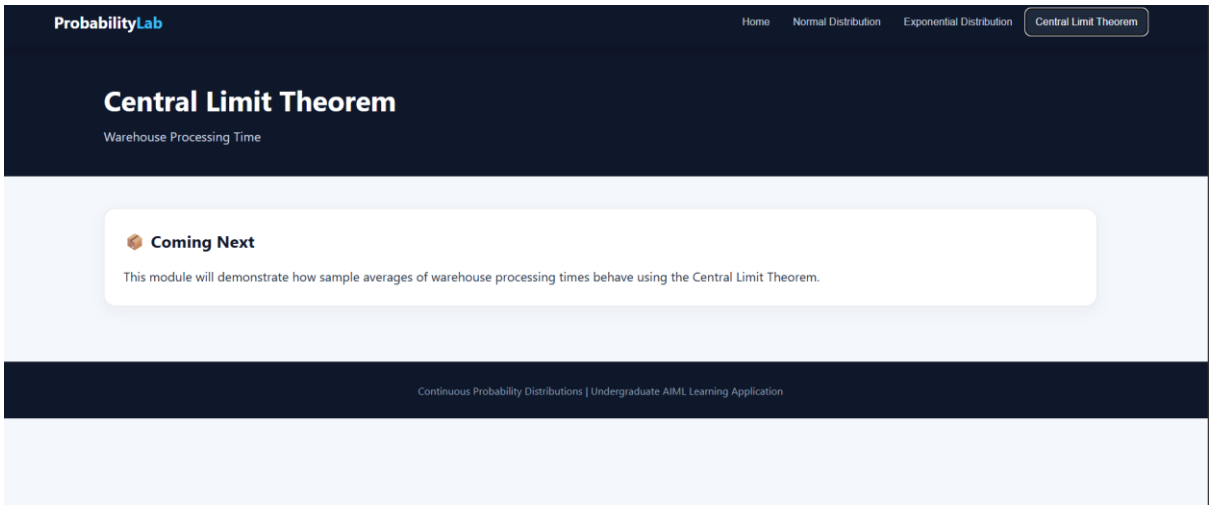
Why CDF? The Normal CDF tells us the probability that a shaft measurement is less than or equal to a particular value. Therefore, subtracting two CDF values gives the probability that the measurement falls between two specification limits.

Exponential Distribution

Cloud Request Inter-arrival Time

Coming Next

This module will model the waiting time between cloud requests using the Exponential Distribution.



Prompt 3 — Add an interactive Exponential Distribution module

Implement the Exponential Distribution — Cloud Request Inter-arrival Time module.

Use an illustrative Poisson request-arrival rate of $\lambda = 12$ requests per minute.

Allow students to change:

- Arrival rate λ in requests per minute
- Time threshold t in seconds

Calculate:

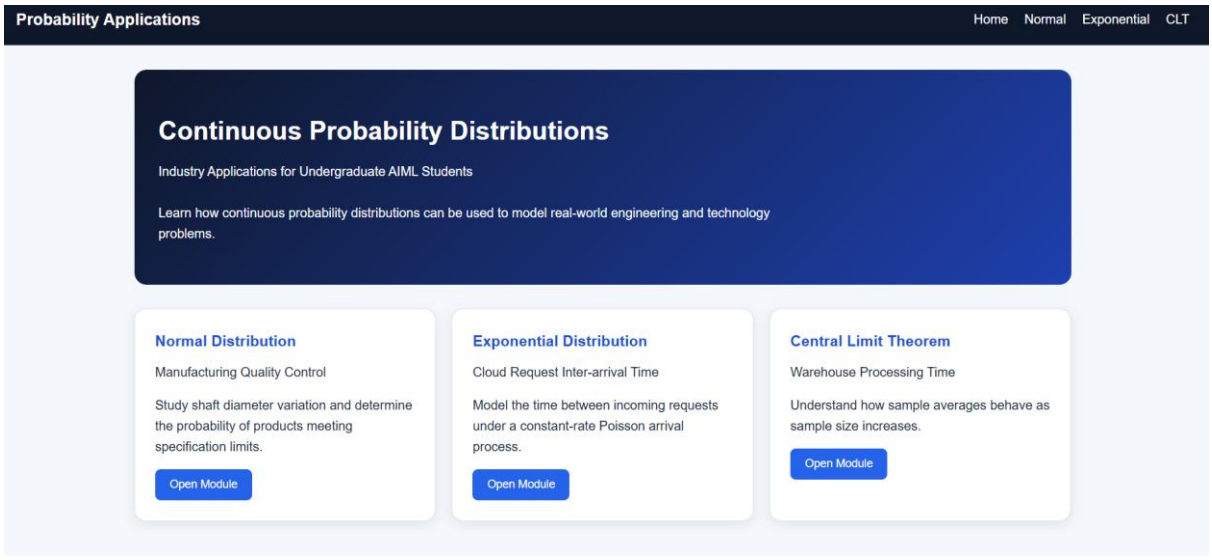
1. Mean inter-arrival time = $1/\lambda$.
2. Variance of inter-arrival time = $1/\lambda^2$, using consistent units.
3. $P(T \leq t) = 1 - \exp(-\lambda t)$.
4. $P(T > t) = \exp(-\lambda t)$.
5. $P(T > t_2 \mid T > t_1)$, where $t_2 \geq t_1$.

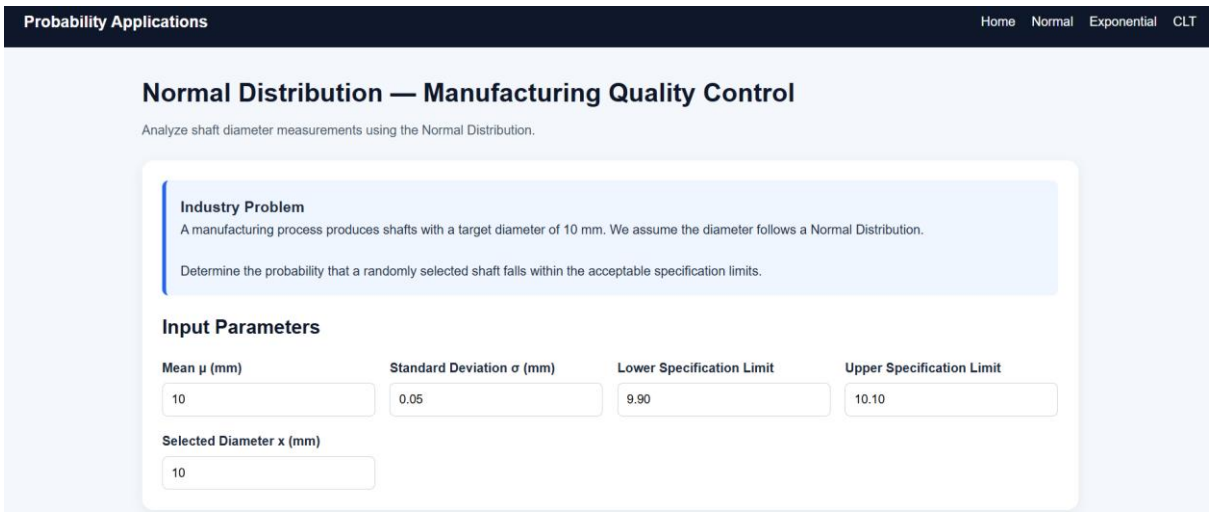
Convert seconds and minutes correctly before calculations. Display the exponential probability density curve and shade the relevant probability region.

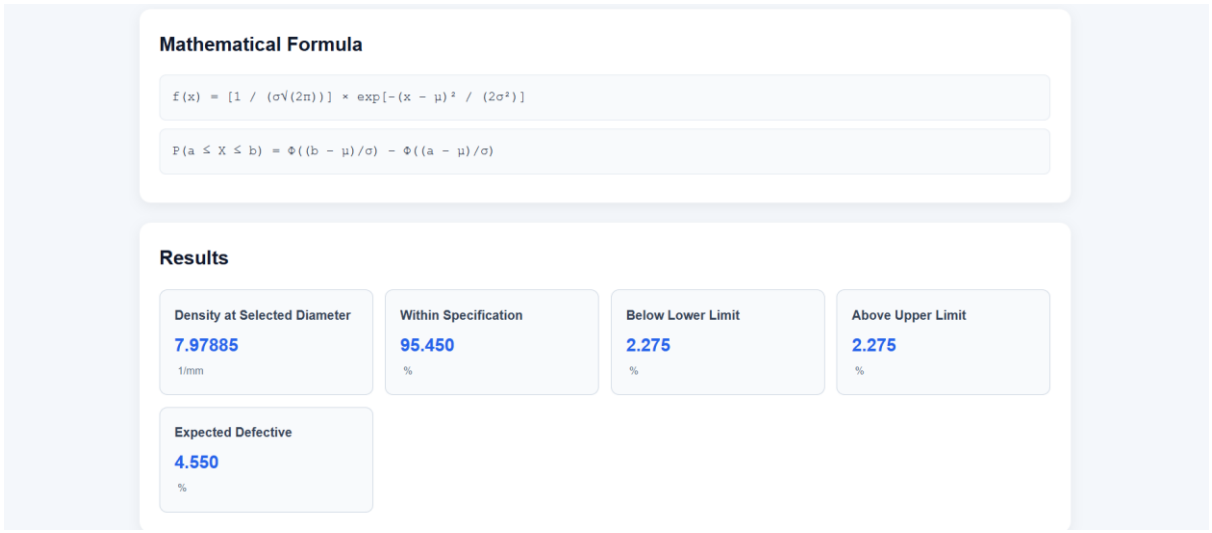
Explain the memoryless property in simple language. Validate all inputs and show units beside every result.

Clearly state that the model assumes a constant-rate Poisson arrival process. Do not describe inter-arrival time as equivalent to server response time.

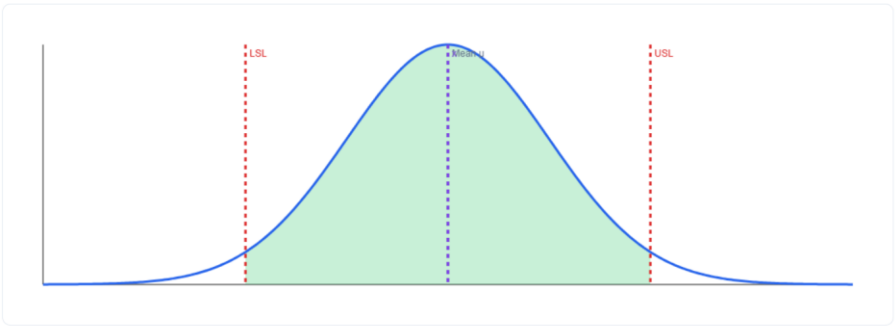
Output:


Probability Applications
Home Normal Exponential CLT
Continuous Probability Distributions
Industry Applications for Undergraduate AIML Students
Learn how continuous probability distributions can be used to model real-world engineering and technology problems.
Normal Distribution
Manufacturing Quality Control
Study shaft diameter variation and determine the probability of products meeting specification limits.
Open Module
Exponential Distribution
Cloud Request Inter-arrival Time
Model the time between incoming requests under a constant-rate Poisson arrival process.
Open Module
Central Limit Theorem
Warehouse Processing Time
Understand how sample averages behave as sample size increases.
Open Module


Probability Applications
Home Normal Exponential CLT
Normal Distribution — Manufacturing Quality Control
Analyze shaft diameter measurements using the Normal Distribution.
Industry Problem
A manufacturing process produces shafts with a target diameter of 10 mm. We assume the diameter follows a Normal Distribution.
Determine the probability that a randomly selected shaft falls within the acceptable specification limits.
Input Parameters
Mean μ (mm)
10
Standard Deviation σ (mm)
0.05
Lower Specification Limit
9.90
Upper Specification Limit
10.10
Selected Diameter x (mm)
10


Mathematical Formula
f(x) = [1 / (σ√(2π))] × exp[-(x - μ)² / (2σ²)]
P(a ≤ X ≤ b) = Φ((b - μ)/σ) - Φ((a - μ)/σ)
Results
Density at Selected Diameter
7.97885
1/mm
Within Specification
95.450
%
Below Lower Limit
2.275
%
Above Upper Limit
2.275
%
Expected Defective
4.550
%

Normal Distribution Curve



Interpretation

95.45% of shafts are expected to fall within the acceptable range of 9.90 mm to 10.10 mm. Approximately 4.55% are expected to be defective because they fall outside the specification limits.

Exponential Distribution — Cloud Request Inter-arrival Time

Model the time between consecutive incoming requests using an Exponential Distribution.

Model Assumption:
This model assumes that cloud requests arrive according to a **constant-rate Poisson arrival process**. Therefore, the time between consecutive requests is modeled using an Exponential Distribution.
Important: Inter-arrival time is the time between arrivals. It is not the same as server response time or processing time.

Industry Problem

Suppose a cloud application receives requests at an average rate of 12 requests per minute.
We want to study how long we wait between consecutive incoming requests.

Input Parameters

Arrival Rate λ requests/minute	Time Threshold t_1 seconds	Time Threshold t_2 seconds
12	3	5
Average number of requests arriving per minute.	Used for $P(T \le t_1)$, $P(T > t_1)$, and conditional probability.	Must satisfy $t_2 \ge t_1$.

Exponential Distribution Formula

If T represents the inter-arrival time:

$$f(t) = \lambda e^{-\lambda t}, \quad t \ge 0$$

$$\text{Mean} = E(T) = 1 / \lambda$$

$$\text{Variance} = \text{Var}(T) = 1 / \lambda^2$$

$$P(T \le t) = 1 - e^{-\lambda t}$$

$$P(T > t) = e^{-\lambda t}$$

$$P(T > t_2 \mid T > t_1) = e^{-\lambda (t_2 - t_1)} \quad \text{for } t_2 \ge t_1$$

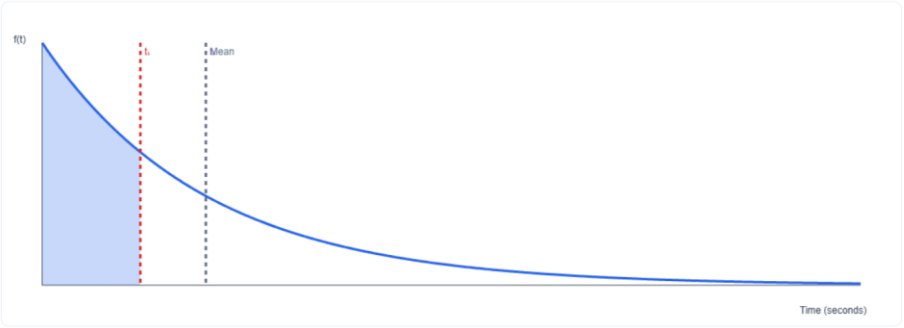
Calculated Results

Mean Inter-arrival Time	Variance	$P(T \le t_1)$	$P(T > t_1)$
5.000 seconds (0.08333 minutes)	25.000 seconds²	45.119 %	54.881 %

$P(T > t_2 \mid T > t_1)$: 67.032 %

Exponential Probability Density Curve

The shaded region represents $P(T \le t_1)$.



Memoryless Property

The Exponential Distribution is called **memoryless** because the probability of waiting an additional amount of time does not depend on how long you have already waited.

For example, if no request has arrived for 3 seconds, the probability of waiting another 2 seconds is the same as it would have been at the beginning.

$$P(T > t_2 \mid T > t_1) = P(T > t_2 - t_1)$$

Interpretation

At an arrival rate of 12.00 requests/minute, the average inter-arrival time is approximately 5.00 seconds. The probability that the next request arrives within 3.00 seconds is 45.12%. The probability of waiting more than 3.00 seconds is 54.88%. If 3.00 seconds have already passed without a request, the probability of waiting beyond 5.00 seconds is 67.03% from that point.

Central Limit Theorem — Warehouse Processing Time

Understand how sample averages behave as sample size increases.

Coming Next

The Central Limit Theorem interactive module will be implemented in the next step.